



Assignment 07: Exploring Lexical Retrieval

handed out: 18 June, 14:00
to be submitted by: 25 June, 12:00

In this assignment, we are going to combine two datasets in order to get an impression whether the retrieval time of lexical items in Chinese, which was shown by Tsang (2018) to be influenced by the complexity of characters, could additionally be influenced by affective ratings as compiled by Mohammad (2018) for English vocabulary items (making the bold assumption that these ratings will transfer between cultures).

Task 1: Joining the Datasets

Load the contents of the Chinese lexical decision dataset `Tsang-2018-LexicalDecision.tsv` and of the affective ratings dataset `Mohammad-2018-AffectiveRatings.tsv` into Pandas DataFrame objects. The two datasets involve data for different languages, but they are both mapped to the concept list items indexed by the Concepticon (<https://concepticon.clld.org/>). Merge the information about Concepticon concepts for which there is data in both datasets into a single DataFrame object, discarding all concepts for which either affective ratings data or lexical decision data is missing.

Task 2: Understanding the Joint Distribution

- In order to get an impression of the structure of the data, group the data points into quintiles along each of the three main variables: number of strokes in the Chinese characters (column `CHINESE_STROKE`), average arousal values (`ENGLISH_AROUSAL_MEAN`), and reaction times (column `CHINESE_RT_MEAN`).
- Cross-tabulate each of the three combinations of two variables (using one for the row index and the other for the column index), and comment whether the overlapping quintile counts reveal anything interesting.
- Cross-tabulate the counts within the reaction time quintiles (column index) with a hierarchical row index formed by the two potential predictor variables (stroke count and arousal). Is the picture getting clearer?

Task 3: Arousal and Reaction Times by Number of Strokes

- Group the data by the number of strokes in the Chinese characters, then compute the average arousal values and reaction times for each group of characters (hint: this can be done using a pivot table).
- Scale the arousal values by a factor of 1000, and plot the result in a grouped barplot ordered by the stroke count. Is the connection between stroke count and lexical retrieval times immediately obvious, and does it look like the affective ratings data might provide independent evidence?

Task 4: Bringing Frequencies into the Picture

In order to determine the potential influence of frequency on lexical retrieval, we are going to use Chinese token frequency counts derived from the Open Subtitles Corpus.

- Import `cmn-opensubtitles-freq.tsv`, transform the raw frequencies in the second column to log frequencies, and merge the result into your existing data frame based on the Chinese lemmas.
- Explore the potential link between log frequency and reaction time through a bivariate scatter plot with marginals. Is there an obvious connection, and how could we explain it?

Task 5: Correlation by Frequency Band

Finally, we investigate whether the correlation we were originally looking for might exist in certain parts of the frequency band (as a cheap way of conditioning on frequency).

- a) Create a grouping defined by separating the concepts into five buckets by their log frequency.
- b) Create a function which measures the correlation of the original two predictor variables with the Chinese reaction times. (Hint: the quantile analysis example is similar in structure)
- c) Use the function within the split-apply-combine pattern in order to see whether a correlation appears on either side of the frequency spectrum.